

Family Health Agent — Round 3 Build Plan

Oct 3, 2026 · @Eshwar

Decisions so far

Seven decisions are locked; six are still open and marked in the workflow below.
Submission closes Sunday 4 Oct, 11:59 PM IST.

Locked

Area	Decision
Form	Standalone app. WhatsApp is the push channel for updates to the patient, responsible person and family group.
Brain	One AI model (Claude or Gemini) running on our system prompt. It makes every decision itself.
Authority	Responsible person (RP) has veto, used only when RP and patient clash. RP otherwise acts for the patient's good.
Money	A wallet: RP sets a limit once and tops it up when comfortable. Demo wallet is ₹50,000. Agent alerts RP on any major spend (an expensive medicine or a costly prescription).
Call chain	Patient first, RP kept in the loop. Then family ranked by ability to make a sound decision, not by age. RP sets the ranking at onboarding.
Scope	End to end: cab and auto included. Ambulance only in severe emergencies.
Language	Any major Indian language. Translation and transcription happen behind the curtain, but only through Gnani: a teammate runs the real audio through Gnani's API and pastes back exactly what it returns. We never write a transcript or translation ourselves.

Open (answer these before we write the system prompt)

1. **Wait time** before the agent moves down the call chain. Proposed: emergency 30 seconds, urgent 3 minutes, routine 15 minutes.
2. **Major spend threshold** as a number. Proposed: any single spend over ₹5,000, or any spend that would take the wallet below 20% of its limit.
3. **Assessing tiers**. Proposed in the workflow below: rules plus context, never a diagnosis.
4. **Cab, auto and clinic booking** have no partner rail. See the rails section for options.
5. **Clash between patient and RP**. Parked for now; RP wins by default.
6. **The recording scenario**. Parked until the workflow is final.

The workflow, rewritten

The agent runs ten stages, from Onboarding to Closing. Every failure moves to the next option, holds, or hands back to a human; none stops silently. The agent records and routes. It never diagnoses, never doses and never substitutes a medicine.

Roles

- **Patient:** the family member who needs care.
- **Responsible person (RP):** manages the family's health and owns the wallet. Holds the veto when they and the patient disagree.
- **Call chain:** patient first, then RP, then other family ranked by the RP for sound judgement.
- **Family group:** a WhatsApp group that gets status updates only.

Three triggers, each with its own severity

Trigger	Routine	Urgent	Critical
Medication	A common medicine runs out (e.g. an antibiotic course)	A chronic medicine the patient must not miss (diabetes, BP), flagged at onboarding	A must-not-miss medicine with no stock anywhere nearby
Medical test	A one-off test (e.g. a cough check)	A recurring test that is due (e.g. diabetes monitoring), or a test needed before a prescription	A test the doctor asked for within hours
Emergency	A minor illness (e.g. flu) needing a clinic visit	Symptoms on the family's red-flag list	Severe: ambulance plus hospital

Stages

0. **Onboarding (day one).** The RP adds family members and the ranked call chain. They upload prescriptions and past reports, and list known conditions and current medicines. They set the red-flag list with the family doctor. They add two known

clinics, chemists and labs each, in ranked order, plus a preferred language per person. The RP sets the wallet limit (₹50,000 in the demo) and the major-spend threshold. Each adult consents to what is shared, and with whom.

1. **Idle.** The agent watches. It knows refill dates (pill count from the prescription) and recurring test dates. It waits for an SOS, a voice note or a message.
2. **Triggered.** Three real sources only: (a) patient presses SOS or sends a voice note, (b) RP or family messages the agent, (c) the clock hits a refill or test date. The agent tags the trigger type and a first severity.
3. **Reaching.** The agent contacts the patient first and tells the RP in parallel. No answer within the wait time → RP → next person in the chain. The RP may add the patient to a conference, or explain the problem themselves.
 - *Unhappy:* nobody answers → move down the chain instead of waiting. Critical severity → call the patient and RP at the same time.
4. **Listening.** The person describes the problem in their own language, by voice note or call. Gnani turns the audio into a transcript, which is stored word for word. The agent writes a short summary.
 - *Unhappy:* patient can't describe it → whoever is present speaks, and the agent marks the account as secondhand for the doctor.
5. **Assessing.** The agent asks follow-up questions against known conditions and current medicines, then gives a severity tier with a one-line reason (see below). It asks the patient whether to act. The RP is told, and can overrule.
 - *Unhappy:* it can't tell → treat it as urgent and ask a human; never guess downward. Critical and nobody reachable → call the ambulance anyway, then keep calling down the chain.
6. **Acting.** The agent does one or more of: book a clinic slot or test, order the medicine, book an auto or cab, pay from the wallet.
 - *Clinic won't pick up* → redial, try the second known clinic, then report the dead end.
 - *No slot* → next available, another known doctor, or walk-in. It says which and why.
 - *Medicine out of stock* → local chemist, then the next. Never substitutes; a substitute goes back to a human.
 - *Area not serviceable for delivery* → nearest chemist pickup, with an auto booked for a family member.
7. **Checking in** (runs alongside stages 3–9). Status goes to the family group. Spend so far and full details go to the RP, and every step is logged. The agent asks a human only for: a spend above the threshold, a new doctor, a new medicine, or a low wallet.

- *Unhappy*: nobody answers on a spend → hold the booking unpaid rather than cancel it.

8. **Handing over.** At the clinic, the agent sends the doctor the record: summary, last reports, current medicines and the trend. The doctor conversation is recorded with consent, transcribed by Gnani and summarised.

- *Unhappy*: wrong specialist → the agent carries the same record into a new booking.

9. **Closing.** The slot was confirmed by name, the auto or cab arrived, the medicine was delivered. Any new prescription updates the medicine list and refill dates. The family is told who did what, the log is closed, and the agent returns to Idle.

Assessing: one step beyond rules

The agent weighs four things, then picks a tier. It cites which factors it used, and never names a condition.

Factor	What the agent checks
Red flags	Does what was said match the family's red-flag list from onboarding?
Change from baseline	Is this new for this patient, or getting worse? It checks how often it has happened recently and what the last reports showed.
Medicine link	Does it touch a known condition, or a medicine the patient takes?
Context	Time of day, who is home and reachable, distance to the clinic, wallet left.

Rule: if the factors disagree, or the agent is unsure, it goes up one tier. It asks the patient to confirm; the RP can overrule. *Proposed, needs your sign-off.*

Where the three rails fit

Gnani is the agent's ears and voice, Pine Labs is the wallet, and Delhivery moves the medicine. Cab, auto, clinic booking and ambulance have no partner rail, so they need a real outside source or an imagined capability.

Real endpoints, by stage

Stage	Rail	Endpoint	What it does for us
4 Listening, 8 Handing over	Gnani (Vachana STT)	POST <code>https://api.vachana.ai/stt/v3</code> with <code>language_code</code> (e.g. <code>hi-IN</code> , <code>te-IN</code>)	Patient's voice note or the doctor conversation → native-script transcript. Covers Hindi, Tamil, Telugu, Bengali, Marathi, Gujarati, Kannada, Malayalam, Punjabi, Odia and Indian English.
3 Reaching, 5 Assessing	Gnani (TTS)	Text-to-speech endpoint in the Gnani docs (confirm exact path at docs.gnani.ai)	The agent speaks its follow-up questions to the patient in their language.
0 Onboarding	Pine Labs (UPI Reserve Pay)	POST <code>/ps/public/subscriptions/sbmd</code>	RP blocks ₹50,000 in their own UPI account. The money stays in their bank until spent.
6 Acting	Pine Labs (UPI Reserve Pay)	POST <code>/ps/api/v1/public/subscriptions/{reserve_pay_sub_id}/presentations</code>	Debits one purchase against the blocked amount.
6 Acting	Pine Labs (Payment Links)	POST <code>/api/pay/v1/paymentlink</code> , then GET <code>/api/pay/v1/paymentlink/{payment_link_id}</code>	Fallback when a spend needs the RP's tap: send a link, check if it is paid. Cancel with PUT <code>.../{payment_link_id}/cancel</code> .
7 Checking in	Pine Labs	GET <code>/ps/public/subscriptions/sbmd/{reserve_pay_sub_id}</code>	Wallet left, shown to the RP at each stage.
6 Acting	Delhivery	GET <code>/c/api/pin-codes/json/?filter_codes=<pin></code>	Can Delhivery deliver to the patient's pincode?
6 Acting	Delhivery	POST <code>/api/cmu/create.json</code>	Creates the shipment from the chemist (a registered pickup location) to the patient; returns a waybill.
6 Acting	Delhivery	Pickup Request Creation API (path in the Delhivery docs, under Pickup Scheduling)	Asks a rider to collect from the chemist.
9 Closing	Delhivery	GET <code>/api/v1/packages/json/?waybill=<id></code>	Confirms delivery.

All Pine Labs paths use the sandbox base `pluraluat.v2.pinepg.in` and a Bearer token from `POST /api/auth/v1/token`. Delhivery uses `staging-express.delhivery.com` with `Authorization: Token <key>`.

Wallet choice. I recommend UPI Reserve Pay over Pine Labs' Brand Wallet (a prepaid wallet you load money into). Reserve Pay tells a better trust story: the RP's money never leaves their account until the agent spends it. Brand Wallet is the backup if the Reserve Pay flow doesn't fit.

The gaps

Need	Real source today	Recommendation
Auto or cab	Namma Yatri runs on the open Beckn / ONDC mobility protocol (search, select, confirm, status).	Use Beckn calls, with the curtain replying exactly as the spec says. Email the organisers tonight to confirm non-partner open protocols are allowed. If not, use imagined capability #3.
Clinic slot	The clinic receptionist on a phone call.	The agent calls; a teammate plays the receptionist; the audio goes through Gnani STT.
Ambulance	108 by phone.	Same as the clinic call. Critical tier only.
Medicine stock check	Calling the chemist.	Phone call through Gnani, or imagined capability #2.
Medical records	Uploaded by the RP at onboarding.	No API needed for the demo.

Imagined capabilities (3 allowed)

1. **Gnani: speaker check on a call.** `POST /voice/v1/speaker-verify` (imagined) returns whether the voice matches the enrolled patient. It makes the "secondhand account" flag automatic. Gnani already sells voice biometrics.
2. **Pine Labs: chemist stock near a pincode.** `GET /merchants/v1/pharmacy/stock?pin=<pin>&sku=<drug>` (imagined). Pine Labs' in-store payment machines at chemists already know merchant category and location, and billing integrations hold item-level sales. This fixes "out of stock, try the next chemist" without blind calls.
3. **Delhivery: same-day pharmacy-to-home pickup.** `POST /hyperlocal/v1/orders` (imagined) with a 4-hour promise. Delhivery already holds rider networks, pincode serviceability and pharma clients. This fixes the fact that its express parcel network is next-day, too slow for an urgent refill.

If the organisers reject Beckn for cabs, capability #1 is the one to swap for a cab capability.

Rail readiness, first take

- **Pine Labs, 9/10.** It publishes an agent toolkit and an MCP server that expose its payment APIs as agent tools.
- **Gnani, 7/10.** Strong Indian-language STT and TTS APIs, but placing calls and handling a call chain isn't in the public speech API.
- **Delhivery, 5/10.** It can serve a chemist only once the chemist is registered as a pickup location. Its express network is built for next-day parcels, not same-day medicine.

Sources: [Gnani STT API](#), [gnani-vachana SDK](#), [Pine Labs API reference](#), [Pine Labs payment links](#), [Pine Labs agent toolkit](#), [Pine Labs MCP server](#), [Delhivery order creation](#), [Delhivery pincode and tracking endpoints](#), [Beckn Mobility MCP server](#)

Build plan

Build a "curtain console" web app with Claude Code, not a mobile simulator. It is one browser page that shows the agent, the curtain and the phones side by side. That is exactly what the recording must show. Phone frames inside the page give you the mobile look without an emulator.

Why you need an API key

Your custom app has to call the model from code, and a claude.ai subscription doesn't include that. Pick one tonight:

- **Gemini API key** (free, about 2 minutes at Google AI Studio). The brief explicitly allows Gemini.
- **Claude API key** (about 10 minutes at console.anthropic.com, roughly \$5 of credit covers the demo).

Without either key, the app can't run on its own, and you're on plan B below.

What the console has

Pane	What it shows	Real or curtain
Curtain (left)	Buttons to feed inputs: upload a voice note, advance the clock, send an SOS. Below them, a queue of pending rail calls with a paste box for each response.	Curtain

Pane	What it shows	Real or curtain
Agent log (centre)	Every model step: the tool call, the request, the response, and a decision card.	Real model
Phones (right)	Two WhatsApp-style frames, patient and RP, plus the family group thread.	Rendered

How the agent loop works

1. The system prompt plus the onboarding data (family, medicines, clinics, wallet) go to the model with a list of tools.
2. The model calls a tool. Three kinds:
 - **Real, run by the app:** `gnani_stt` and `gnani_tts` call Gnani directly with your key; `log_decision` and `wallet_ledger` write to a Google Sheet through an Apps Script webhook.
 - **Curtain:** `pinelabs_*`, `delhivery_*`, `beckn_mobility_*`, `place_call`. The app pauses and shows the request; a teammate pastes the response from the docs.
 - **Messages:** `send_whatsapp` renders in the phone frame. Real WhatsApp through the Twilio sandbox is a stretch goal only.
3. The model must call `log_decision` at every choice, with: time, input, source, decision, rule ID, message sent, recipient, connector. That table is the Part 1 decision list, produced by the run itself.
4. Every system-prompt rule carries an ID (R1, R2...) so each decision cites one.

Plan B, no build

A Claude project or Gemini Gem holding the system prompt. The curtain pastes inputs and responses into the chat by hand. Ready in 2 hours, but it looks weaker on video, and you build the decision table by hand. Switch at 1 PM Sunday if the console isn't running.

Timeline to submission

When	Who	What
Sat 10:30 PM – 1 AM	You	Close the six open decisions. Get the API key. Run one real voice note through Gnani to prove the key works. Email the organisers about Beckn.
Sat 10:30 PM – 1 AM	Teammate	Write the system prompt v1 from the workflow, with rule IDs.
Sun 8 AM – 1 PM	You + Claude Code	Build the console. Agent loop first, then the curtain pane, then phones, then the Sheet.
Sun 8 AM – 1 PM	Teammates	Screen mockups, endpoint request/response doc, rail scores, Part 2 answers, landing page.
Sun 1 PM	All	Go or no-go on the console. Plan B if no.
Sun 1 – 4 PM	All	Two dry runs, fix the prompt, then record two or three takes.
Sun 4 – 10 PM	All	Edit the video to 5 minutes, write the decision list from the log, assemble the submission.
Sun 10 PM – 11:59 PM	All	Buffer. Submit by 11 PM.

Running the curtain

Split the curtain into three jobs so nobody improvises. With two people, one person covers People and Rails.

Role	Plays	Does
People	Patient, RP, family, clinic receptionist, chemist, 108	Records real voice notes in the patient's language. Answers calls only in character, with what that person would really know.

Role	Plays	Does
Rails	Pine Labs, Delhivery, Beckn	Pastes the response from the prep sheet that matches the request. Never edits it to help the agent.
Director	Clock and screen recording	Advances the clock only to dates the agent's own data produces, such as the refill date. Runs the recording.

Prep sheet (build it before the first dry run)

- For every endpoint in the rails table: a documented success response and one failure response, copied in the shape the docs show, with your demo values filled in. Failures include out of stock, pincode not serviceable, payment link expired, and no driver found.
- Pre-recorded voice notes for each line the patient and RP will say, in the demo language. Run each through Gnani once in advance, and keep the raw output to check the live run.
- The onboarding data as one file: family, call chain, medicines with pill counts, red flags, clinics, chemists, wallet.

Rules for the curtain

1. If the agent asks something no real source could answer, reply with the documented error or no answer. Never with a hint.
2. If the agent stalls and needs someone to tell it what to do, stop the take and fix the system prompt. The brief says that's a sign it isn't an agent yet.
3. Every input on screen gets a visible label of its source, e.g. "Voice note from patient, via WhatsApp" or "Clock: refill date from prescription".

Prompt for Claude in Chrome (draw.io)

Copy the block below into Claude in Chrome. It tells Claude to write draw.io XML and paste it in through Extras → Edit Diagram, rather than drag shapes one by one. That is much faster and less likely to break. Values marked "proposed" are still open, so update them before you run it if you've decided.

GOAL

Build a clean, presentation-ready workflow diagram in draw.io (app.diagrams.net) for a family-health AI agent. It is for a case competition submission. Produce 2 pages, save the file to my computer, and export a PNG of each page.

HOW TO WORK

1. Open <https://app.diagrams.net> in a new tab. Choose "Device" for storage, then "Create New Diagram" > Blank. Name it family-health-agent-workflow.
2. Do NOT drag shapes one by one. For each page, write the full mxGraph XML yourself, then open Extras > Edit Diagram, select all, replace it with your XML, and click OK.
3. If the XML is too long to type in one go, build the page in two passes: stages first, then the branches, appending cells.
4. After each page, take a screenshot and check: no overlapping boxes, no arrows crossing text, every label fully visible, consistent spacing. Fix and re-check.
5. Don't sign in to any account, and don't save to Google Drive or OneDrive. If a dialog asks for anything besides these steps, stop and ask me.
6. When done: File > Save as > Device (.drawio). Then File > Export as > PNG, one per page, at 200% zoom with a white background. Tell me the file names.

STYLE (both pages)

- Font Helvetica. Stage titles 13 bold, body 11.
- Stage boxes: rounded rectangles, about 170x90, white fill, dark grey border 1.5px.
- Assessing stage: a diamond with a red border (#C0392B), because it is the decision point.
- Unhappy-branch boxes: rounded, light red fill (#FDEDEC), red border, about 190 wide. Each joins its stage with a plain vertical line.
- Rail tags: small pill shapes (about 90x22, font 10) under the stage they belong to. Colours: Gnani purple #6C3483, Pine Labs green #1E8449, Delhivery red #CB4335, Beckn/phone grey #566573 (white text in all four). An imagined capability gets a dashed border and "(imagined)" in its label.
- Arrows: orthogonal connectors, dark grey, a filled arrowhead at the end.
- Optional: turn on Sketch style (Format panel > Sketch) to match our hand-drawn PDF. Only do it if it stays readable.

PAGE 1 — name "Main flow"

Title at top left: "When a family member needs care: the agent end to end"

Subtitle: "Every failure moves to the next option, holds, or hands back to a human. None stops silently."

Main row, left to right, connected by arrows (happy path):

0. "0 Onboarding" – "RP adds family, ranked call chain, prescriptions, reports, red flags, clinics/chemists/labs, language, wallet ₹50,000, consent"
1. "1 Idle" – "Watches refill and test dates. Waits for SOS, voice note or message"
2. "2 Triggered" – "Type: medication / test / emergency. First severity: routine / urgent / critical"
3. "3 Reaching" – "Calls patient first, tells RP in parallel. Then down the ranked chain"
4. "4 Listening" – "Person describes it in their own language. Transcribed word for word, then summarised"
5. "5 Assessing" (diamond) – "Follow-up questions. Severity tier + reason. Patient confirms, RP can veto"
6. "6 Acting" – "Books clinic/test, orders medicine, books auto/cab, pays from wallet"
7. "7 Checking in" – "Status to family group. Spend + details to RP. Asks a human only for big spend, new doctor, new medicine, low wallet"
8. "8 Handing over" – "Record goes to the doctor: summary, reports, medicines, trend. Doctor talk recorded and summarised"
9. "9 Closing" – "Slot confirmed by name, cab arrived, medicine delivered. New prescription updates refill dates. Family told who did what"

From the Assessing diamond, two labelled exits: "Routine / Urgent" to Acting, and "Critical" to Acting via a short red arrow labelled "act now, ask after". A dashed arrow from 9 Closing back to 1 Idle, routed above the row, labelled "back to watching".

A thin dashed bracket under stages 3–9 labelled "Checking in runs alongside these stages".

Above 2 Triggered: three small input boxes with arrows into it:

"Patient SOS or voice note (WhatsApp)", "RP or family message", "Clock: refill or test date".

Rail tags under the stages:

- 3 Reaching: "Gnani TTS"
- 4 Listening: "Gnani STT"; "Gnani speaker check (imagined)"
- 5 Assessing: "Gnani TTS"
- 6 Acting: "Pine Labs Reserve Pay debit"; "Pine Labs payment link"; "Delhivery pincode check"; "Delhivery shipment + pickup"; "Beckn auto/cab"; "Phone: clinic, 108"; "Pine Labs chemist stock (imagined)"; "Delhivery same-day (imagined)"
- 7 Checking in: "Pine Labs wallet balance"
- 8 Handing over: "Gnani STT"

- 9 Closing: "Delhivery tracking"
- 0 Onboarding: "Pine Labs Reserve Pay block"

Unhappy branches, below the rail tags of their stage:

- 3 Reaching: "Nobody answers – moves to the next person in the chain instead of waiting (wait: 30s critical / 3 min urgent / 15 min routine, proposed)"
- 4 Listening: "Patient can't describe it – whoever is present speaks. Flagged to the doctor as secondhand"
- 5 Assessing: "Can't tell – treats it as urgent and asks a human. Never guesses downward"
- 5 Assessing: "Critical, nobody reachable – calls the ambulance anyway, then keeps calling down the chain"
- 6 Acting: "Clinic won't pick up – redials, tries the second clinic, reports the dead end"
- 6 Acting: "No slot – next available, another known doctor, or walk-in. Says which and why"
- 6 Acting: "Medicine out of stock – next chemist. Never substitutes; that goes back to a human"
- 6 Acting: "Pincode not serviceable – nearest chemist pickup, auto booked for a family member"
- 7 Checking in: "Spend above threshold (over ₹5,000 or wallet below 20%, proposed) – stops and asks the RP. No answer: holds the booking unpaid, doesn't cancel"
- 8 Handing over: "Wrong specialist – carries the same record into a new booking"

Stack several branches under one stage vertically, 16px apart.

Footer line: "The agent records and routes. It never diagnoses, doses, or substitutes a medicine."

Legend at bottom right: one sample each of stage box, decision diamond, unhappy branch, the four rail colours, and the imagined dashed pill.

PAGE 2 – name "Assessing and authority"

Title: "How the agent decides how serious it is, and who has the final say"

Left column – four input boxes stacked vertically, each with an arrow into one diamond:

- "Red flags – matches the family's red-flag list from onboarding?"
- "Change from baseline – new for this patient, or getting worse?"
- "Medicine link – touches a known condition or current medicine?"
- "Context – time, who is reachable, distance to clinic, wallet left"

Centre: diamond "Pick tier + one-line reason". Note under it: "Unsure or factors disagree → go up one tier".

From the diamond, four exits to four boxes: "Routine", "Urgent", "Critical", "Can't tell → treat as Urgent, ask a human".

Then a flow on the right:

"Ask patient: act on this?" → diamond "Patient and RP agree?"

- yes → "Act (stage 6)"

- no → "RP decides (veto)" → "Act" or "Stop and log"

The Critical box also has a direct red arrow to "Act now: ambulance + call chain in parallel", which skips consent.

Bottom: a table with 4 columns (Trigger / Routine / Urgent / Critical) and 3 rows:

- Medication | Common medicine runs out | Must-not-miss chronic medicine (e.g. diabetes, BP) | Must-not-miss medicine, no stock nearby

- Medical test | One-off test | Recurring test due, or a test needed for a prescription | Doctor wants it within hours

- Emergency | Minor illness, clinic visit | Symptom on the red-flag list | Severe: ambulance + hospital

FINISH

Re-check both pages for overlaps and cut-off text, save, export, and report the file names plus anything you couldn't do.